

Finite Element Method

Strong and Weak Forms in 2D and 3D

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The flux vector $\vec{q}$ can be expressed in terms of two components: the component tangential to the boundary q_t and the component normal to the boundary q_n . The tangential component q_t does not contribute to the heat entering or exiting the control volume. Recall that

$$\vec{q} = q_x \vec{i} + q_y \vec{j}, \quad \vec{n} = n_x \vec{i} + n_y \vec{j}, \quad n_x^2 + n_y^2 = 1.$$

The normal component q_n is given by the scalar product of the heat flux with the normal to the body:

$$q_n = \vec{q} \cdot \vec{n} = \mathbf{q}^T \mathbf{n} = q_x n_x + q_y n_y. \quad (1)$$

On AD, where $\vec{n} = -\vec{i}$, the heat inflow is $-q_n = -\vec{q} \cdot (-\vec{i}) = q_x$ whereas on BC, where $\vec{n} = \vec{i}$, the heat inflow is $-q_n = -\vec{q} \cdot \vec{i} = -q_x$.

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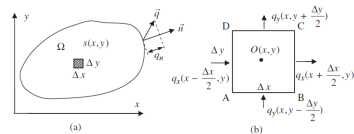


Figure 6.5 Problem definition: (a) domain of a plate with a control volume shaded and (b) heat fluxes in and out of the control volume.

Figure 6.5 Problem definition: (a) domain with control volume shaded and (b) heat fluxes in and out of the control volume.

- The energy balance in the control volume is given by

$$q_x\left(x - \frac{\Delta x}{2}, y\right) \Delta y - q_x\left(x + \frac{\Delta x}{2}, y\right) \Delta y + q_y\left(x, y - \frac{\Delta y}{2}\right) \Delta x - q_y\left(x, y + \frac{\Delta y}{2}\right) \Delta x + s(x, y) \Delta x \Delta y = 0.$$

where the first four terms are the net heat inflow. Dividing by $\Delta x \Delta y$ and recalling the definition of a partial derivative:

$$\lim_{\Delta x \rightarrow 0} \frac{q_x\left(x + \frac{\Delta x}{2}, y\right) - q_x\left(x - \frac{\Delta x}{2}, y\right)}{\Delta x} = \frac{\partial q_x}{\partial x}, \quad \lim_{\Delta y \rightarrow 0} \frac{q_y\left(x, y + \frac{\Delta y}{2}\right) - q_y\left(x, y - \frac{\Delta y}{2}\right)}{\Delta y} = \frac{\partial q_y}{\partial y}.$$

The above energy balance equation (after a change of sign) can then be written as

$$\frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y} - s = 0,$$

or in the vector and matrix forms:

$$(a) \quad \vec{\nabla} \cdot \vec{q} - s = 0 \quad \text{or} \quad \text{div } \vec{q} - s = 0 \quad \text{or} \quad (b) \quad \nabla^T \mathbf{q} - s = 0. \quad (2)$$

- In two dimensions, we have two flux components and two temperature gradient components. For isotropic materials in two dimensions, Fourier's law is given by

$$\vec{q} = -k\vec{\nabla}T \quad \text{or} \quad \mathbf{q} = -k\nabla T, \quad (3)$$

- where $k > 0$. As in one dimension, the minus sign in (3) reflects the fact that heat flows in the direction opposite to the gradient, i.e. from high temperature to low temperature.
- By substituting (3) into (2):

$$k\nabla^2 T + s = 0, \quad (4)$$

where

$$\nabla^2 = \vec{\nabla} \cdot \vec{\nabla} = \nabla^T \nabla = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}.$$

- Equation (4) is called the **Poisson equation** and ∇^2 is called the **Laplacian operator**.

- The flux and the temperature gradient vectors are related by a generalized Fourier's law:

$$\begin{bmatrix} q_x \\ q_y \end{bmatrix} = - \underbrace{\begin{bmatrix} k_{xx} & k_{xy} \\ k_{yx} & k_{yy} \end{bmatrix}}_{\mathbf{D}} \begin{bmatrix} \frac{\partial T}{\partial x} \\ \frac{\partial T}{\partial y} \end{bmatrix}, \quad (5)$$

or in the matrix form:

$$\mathbf{q} = -\mathbf{D}\nabla T, \quad (6)$$

where $\mathbf{D}$ is the conductivity matrix. Substituting the generalized Fourier law (6) into the energy balance equation (2) yields

$$\nabla^T(\mathbf{D}\nabla T) + s = 0. \quad (7)$$

For isotropic materials,

$$\mathbf{D} = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} = k\mathbf{I}. \quad (8)$$

- The strong form for the heat conduction problem in two dimensions is given in the vector form for isotropic materials as

$$\begin{aligned}
 & \text{(a) energy balance: } \vec{\nabla} \cdot \vec{q} - s = 0 \quad \text{on } \Omega, \\
 & \text{(b) Fourier's law: } \vec{q} = -k \vec{\nabla} T \quad \text{on } \Omega, \\
 & \text{(c) natural BC: } q_n = \vec{q} \cdot \vec{n} = \bar{q} \quad \text{on } \Gamma_q, \\
 & \text{(d) essential BC: } T = \bar{T} \quad \text{on } \Gamma_T.
 \end{aligned} \tag{9}$$

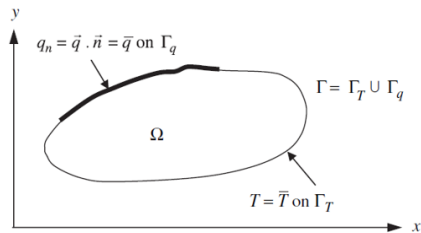
- And in the matrix form

$$\begin{aligned}
 & \text{(a) energy balance: } \nabla^T \mathbf{q} - s = 0 \quad \text{on } \Omega, \\
 & \text{(b) Fourier's law: } \mathbf{q} = -\mathbf{D} \nabla T \quad \text{on } \Omega, \\
 & \text{(c) natural BC: } q_n = \mathbf{q}^T \mathbf{n} = \bar{q} \quad \text{on } \Gamma_q, \\
 & \text{(d) essential BC: } T = \bar{T} \quad \text{on } \Gamma_T.
 \end{aligned} \tag{10}$$

- For isotropic materials,

$$\mathbf{D} = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} = k \mathbf{I}. \tag{11}$$

- κ is the thermal conductivity of the material (W/mK)
- T is the temperature field
- $\mathbf{q}$ is the local heat flux density (W/m²)
- s is the heat generated



Problem domain and boundary conditions.

- To obtain the weak form we follow the same basic procedure as for the one-dimensional problem. We premultiply the two equations by a weight function w and integrate over the problem domain Ω and the natural boundary Γ_q , respectively:

$$(a) \quad \int_{\Omega} w(\vec{\nabla} \cdot \vec{q} - s) d\Omega = 0 \quad \forall w, \quad (b) \quad \int_{\Gamma_q} w(\bar{q} - \vec{q} \cdot \vec{n}) d\Gamma = 0 \quad \forall w. \quad (12)$$

- Applying Green's formula to the first term yields

$$\int_{\Omega} w \vec{\nabla} \cdot \vec{q} d\Omega = \oint_{\Gamma} w \vec{q} \cdot \vec{n} d\Gamma - \int_{\Omega} \vec{\nabla} w \cdot \vec{q} d\Omega \quad \forall w. \quad (13)$$

- Inserting (13) into (12a) yields

$$\int_{\Omega} \vec{\nabla} w \cdot \vec{q} d\Omega = \oint_{\Gamma} w \vec{q} \cdot \vec{n} d\Gamma - \int_{\Omega} w s d\Omega = \int_{\Gamma_q} w \vec{q} \cdot \vec{n} d\Gamma + \int_{\Gamma_T} w \vec{q} \cdot \vec{n} d\Gamma - \int_{\Omega} w s d\Omega. \quad (14)$$

- Substituting (12b) into the integral on Γ_q in (14) yields

$$\int_{\Omega} \vec{\nabla} w \cdot \vec{q} d\Omega = \int_{\Gamma_q} w \bar{q} d\Gamma + \int_{\Gamma_T} w \vec{q} \cdot \vec{n} d\Gamma - \int_{\Omega} w s d\Omega. \quad (15)$$

- It is easy to construct weight functions that vanish on a portion of the boundary, so we set $w = 0$ on the prescribed temperature boundary, i.e. the essential boundary. Therefore, the integral on Γ_T vanishes and the weak form is given by

$$\int_{\Omega} \vec{\nabla} w \cdot \vec{q} d\Omega = \int_{\Gamma_q} w \bar{q} d\Gamma - \int_{\Omega} w s d\Omega \quad \forall w \in U_0. \quad (16)$$

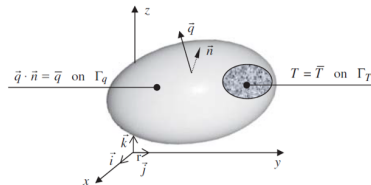
- Expressing (16) in matrix form gives

$$\int_{\Omega} (\nabla w)^T \mathbf{q} \, d\Omega = \int_{\Gamma_q} w \bar{q} \, d\Gamma - \int_{\Omega} w s \, d\Omega \quad \forall w \in U_0. \quad (17)$$

- The above is the weak form for any material, linear or nonlinear.
- To obtain the weak form for linear materials, we substitute Fourier's law into the first term of the above, which yields find $T \in U$ such that:

$$\int_{\Omega} (\nabla w)^T \mathbf{D} \nabla T \, d\Omega = - \int_{\Gamma_q} w \bar{q} \, d\Gamma + \int_{\Omega} w s \, d\Omega \quad \forall w \in U_0. \quad (18)$$

- The extension from two to three dimensions is almost trivial. The difference is not in the structure of the strong and weak form equations, which are identical, but in the definitions of the vectors, gradient, divergence and Laplacian operators.



Problem domain and boundary conditions in three dimensions.

- The progression of dimensionality of the problem domain and its boundary from one-dimensional to three-dimensional problems is summarized in the table

Dimensionality of the problem domain and its boundary.		
Entity	Domain Ω	Boundary Γ
One dimension (1D)	Line segment	Two end points
Two dimensions (2D)	Two-dimensional area	Curve
Three dimensions (3D)	Volume	Surface

- The gradient operator in three dimensions in vector and matrix notations is defined as

$$\vec{\nabla} = \vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z}, \quad \vec{\nabla} \theta = \vec{i} \frac{\partial \theta}{\partial x} + \vec{j} \frac{\partial \theta}{\partial y} + \vec{k} \frac{\partial \theta}{\partial z},$$

$$\nabla = \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix}, \quad \nabla \theta = \begin{bmatrix} \frac{\partial \theta}{\partial x} \\ \frac{\partial \theta}{\partial y} \\ \frac{\partial \theta}{\partial z} \end{bmatrix}.$$

- With the above definitions of vectors and the gradient vector operator, the divergence of the vector field and the Laplacian are

$$\text{div } \vec{q} = \frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y} + \frac{\partial q_z}{\partial z}, \quad \nabla^2 = \vec{\nabla} \cdot \vec{\nabla} = \nabla^T \nabla = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}.$$

- The strong form in vector and matrix notations is identical to that given in Eqs. (9) and (10).
- Note that the Fourier law relating the three components of temperature gradient to the three flux components is defined in terms of a 3×3 symmetric positive definite matrix D :

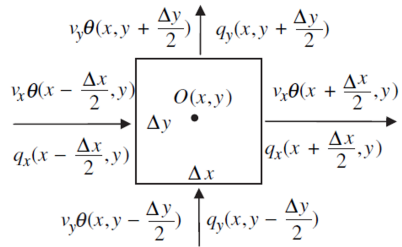
$$\mathbf{D} = \begin{bmatrix} k_{xx} & k_{xy} & k_{xz} \\ k_{yx} & k_{yy} & k_{yz} \\ k_{zx} & k_{zy} & k_{zz} \end{bmatrix}.$$

- The weak form is also identical to that for two-dimensional problems as given in (18).

- The advection-diffusion equations are obtained from a conservation principle, just like heat conduction. The conservation principle states that the species (be it a material, an energy or a state) are conserved in each control volume of area shown in the figure.
- The general form of the steady-state advection-diffusion equation in two dimensions is:

$$\vec{\nabla} \cdot (\theta \vec{v}) + \vec{\nabla} \cdot \vec{q} - s = 0 \quad (19)$$

- $\vec{v} = (v_x, v_y)$ is the velocity vector
- θ is a scalar quantity (such as temperature or concentration)
- $\vec{\nabla} \cdot (\theta \vec{v})$ represents advection
- $\vec{\nabla} \cdot \vec{q}$ represents diffusion



Control volume for advection–diffusion problem.

Control volume for advection–diffusion problem.

- The continuity equation for steady-state problems of incompressible materials states

$$\vec{\nabla} \cdot \vec{v} = 0 \quad \text{or} \quad \nabla^T \mathbf{v} = 0. \quad (20)$$

- Substituting Eq. (20) into Eq. (19) gives

$$\vec{v} \cdot \vec{\nabla} \theta + \vec{\nabla} \cdot \vec{q} - s = 0 \quad \text{or} \quad \mathbf{v}^T \nabla \theta + \nabla^T \mathbf{q} - s = 0. \quad (21)$$

- Assuming that the generalized Fourier's law holds, the conservation of species equation in the matrix form becomes

$$\mathbf{v}^T \nabla \theta - \nabla^T (\mathbf{D} \nabla \theta) - s = 0. \quad (22)$$

For isotropic materials $\mathbf{D} = k\mathbf{I}$ and the conservation equation reduces to

$$\vec{v} \cdot \vec{\nabla} \theta - k \nabla^2 \theta - s = 0 \quad \text{or} \quad \mathbf{v}^T \nabla \theta - k \nabla^2 \theta - s = 0, \quad (23)$$

where ∇^2 is the Laplacian. We consider the usual essential and natural boundary conditions

$$\theta = \bar{\theta} \quad \text{on } \Gamma_\theta, \quad \vec{q} \cdot \vec{n} = \bar{q} \quad \text{on } \Gamma_q, \quad (24)$$

where Γ_θ and Γ_q are complementary.

For a steady-state system, the advection-diffusion equation is:

$$\underbrace{\mathbf{v} \cdot \nabla \theta}_{\text{Advection (Transport by fluid flow)}} = \underbrace{\nabla \cdot (D \nabla \theta)}_{\text{Diffusion (Spreading by molecular transport)}} + \underbrace{S}_{\text{Source/Sink}}$$

This equation states that:

- The **rate of change** of θ due to advection (left-hand side)
- Is **balanced by** diffusion and sources/sinks (right-hand side).

Physical Interpretation

- If advection dominates ($v \gg D$), transport is mostly governed by bulk fluid motion (e.g., large-scale fluid flow).
- If diffusion dominates ($D \gg v$), the material spreads more due to random molecular motion (e.g., slow-moving or stagnant fluids).
- If there is a source term ($S \neq 0$), it means θ is being generated (e.g., heat generation in a reaction) or depleted (e.g., chemical consumption).

Pollutant Dispersion in the Atmosphere

Equation:

$$\mathbf{v} \cdot \nabla C = D \nabla^2 C + S$$

where C is pollutant concentration.

Physical Example:

- A **factory chimney** releases smoke (source term S).
- **Wind (advection)** carries pollutants downwind.
- **Diffusion** spreads the pollutants in different directions.

Semiconductor Heat Dissipation

Equation:

$$\mathbf{v} \cdot \nabla T = k \nabla^2 T + S$$

Physical Example:

- **Microprocessors generate heat** (source term S).
- **Coolant or airflow (advection)** removes heat.
- **Thermal diffusion** spreads heat within the material.

Spread of Airborne Diseases

Equation:

$$\mathbf{v} \cdot \nabla C = D \nabla^2 C + S$$

Physical Example:

- **Coughing or sneezing** releases viral particles.
- **Airflow (advection)** moves the particles.
- **Diffusion** spreads them in a room or an airplane.

Ocean Currents and Climate Models

Equation:

$$\mathbf{v} \cdot \nabla C = D \nabla^2 C$$

Physical Example:

- **Carbon dioxide dissolved in seawater** is transported by ocean currents.
- **Advection** moves CO_2 across the ocean.
- **Diffusion** spreads CO_2 into deeper layers.

Next: Approximations of Trial Solutions for 2D and 3D Problems — Part 1

Thanks for your attention!